CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 7: Transaction Concurrency with Strict 2-Phase Locking

Course: Advanced Database Engineering Laboratory [23CS3506]
Category: Experiments / Practicals | Experiment #7 | Academic Year: 2026-27

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1. Overview & Objective:

Simulate concurrent sessions verifying shared vs exclusive lock acquisition and deadlock timeouts.

2. Practical Tasks & Verification Steps:

1. Open two parallel terminal sessions connected to PostgreSQL transaction engine.
2. In Session 1: BEGIN; SELECT ... FOR SHARE; verify Session 2 can read but cannot write blocking .
3. In Session 2: UPDATE account balance; demonstrate wait state until Session 1 issues COMMIT.
4. Intentionally cause mutual cyclic wait Session 1 waits on 2, Session 2 waits on 1 verifying deadlock abort.

Submission Guidelines & Verification:

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